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A Comparison of OLMo and Qwen2 model families

Feature	OLMo-1B	OLMo-7B	Qwen2-0.5B	Qwen2-1.5B	Qwen2-7B
Hidden Size	2048	4096	896	1536	3584
# Layers	16	32	24	28	28
# Attention Heads	16	32	14	12	28
# Key/Value Heads	16	32	2	2	4
Intermediate Size	8192	11008	4864	8960	18944
Normalization	Non-parametric layer norm		RMSNorm		
Vocabulary Size	50,304		151,936		
Weight Tying	Yes	No	Yes	Yes	No

Table 1: Comparison of OLMo and Qwen2 model configurations

B Similar and dissimilar prompts

D1:
p_A = “ The house at the end of the street was very”, top prediction = “ quiet”
p_B = “ The house at the end of the street was in”, top prediction = “ a”
D2:
p_A = “ He suddenly looked at his watch and realized he was”, top prediction = “ late”
p_B = “ He suddenly looked at his watch and realized he had”, top prediction = “ been”
D3:
p_A = “ And then she picked up the phone to call her”, top prediction = “ mom”
p_B = “ And then she picked up the phone to call him”, top prediction = “.”
S1:
p_A = “ She opened the dusty book and a cloud of mist”
p_B = “ She opened the dusty book and a cloud of dust”
S2:
p_A = “ In the quiet library, students flipped through pages of”
p_B = “ In the quiet library, students flipped through pages in”
S3:
p_A = “ The hiker reached the peak and admired the breathtaking”
p_B = “ The hiker reached the peak and admired the spectacular”

C Details of the 2D slice visualization

We create synthetic activations X in the 2D slice using the formula:

$$X = A + \alpha(B - A) + \beta(C - P),$$

where P is the orthogonal projection of C onto the line containing A and B, and α and β range from -0.25 to 1.25. This allows us to examine a broad area around and between the three reference activations.

To compute the colors for each point:

For each point X in the 2D slice, we compute three distances:

$$d_A = \|F_{p_A}(A) - F_{p_A}(X)\|_2$$
$$d_B = \|F_{p_B}(B) - F_{p_B}(X)\|_2$$
$$d_C = \|F_{p_C}(C) - F_{p_C}(X)\|_2$$

where F_p is defined as in Section 3. We normalize these distances by dividing by their respective maximum values over all points in the slice:

$$\hat{d}_A = \frac{d_A}{\max_X d_A}, \ \hat{d}_B = \frac{d_B}{\max_X d_B}, \ \hat{d}_C = \frac{d_C}{\max_X d_C}$$

The RGB color components for point X are then computed as: $\texttt{Red} = 1 - \hat{d}_A$, $\texttt{Green} = 1 - \hat{d}_B$, $\texttt{Blue} = 1 - \hat{d}_C$.

D More 2D slice plots

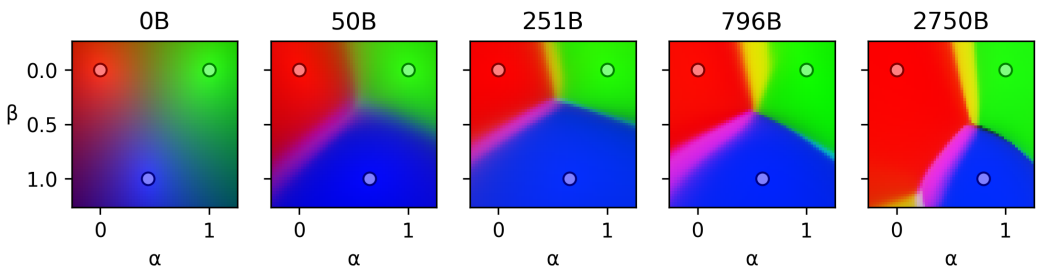

Figure 5

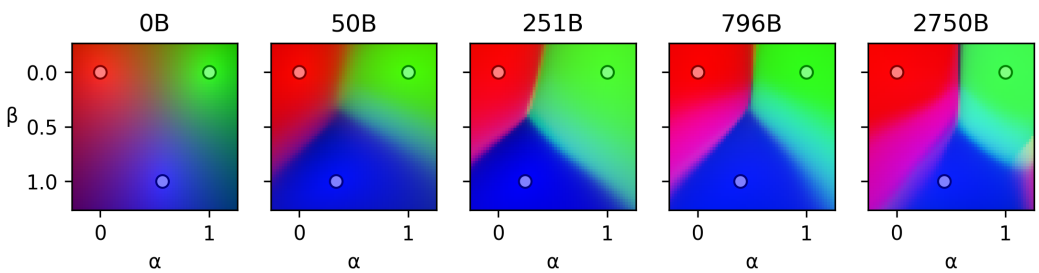

Figure 6

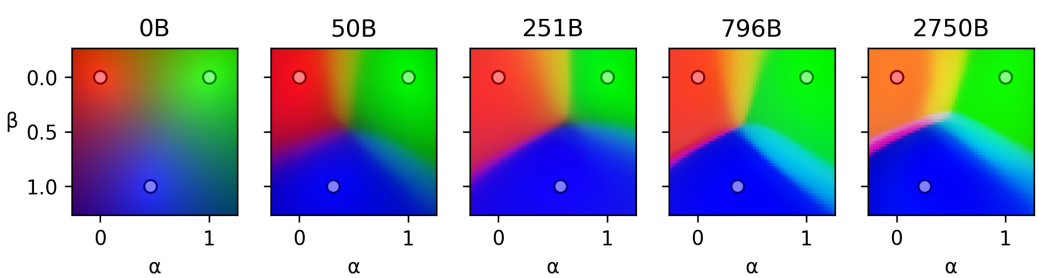

Figure 7

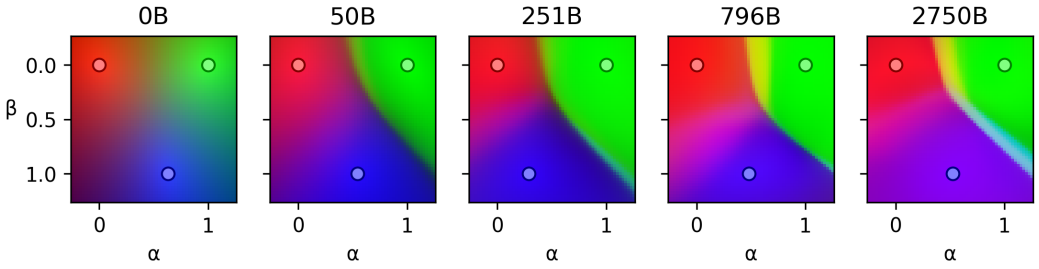

Figure 8

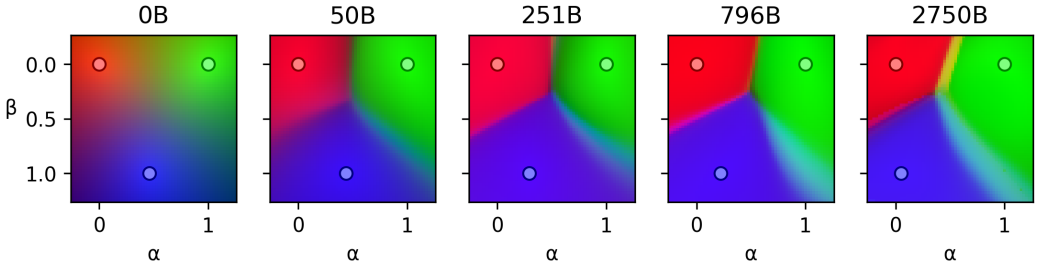

Figure 9

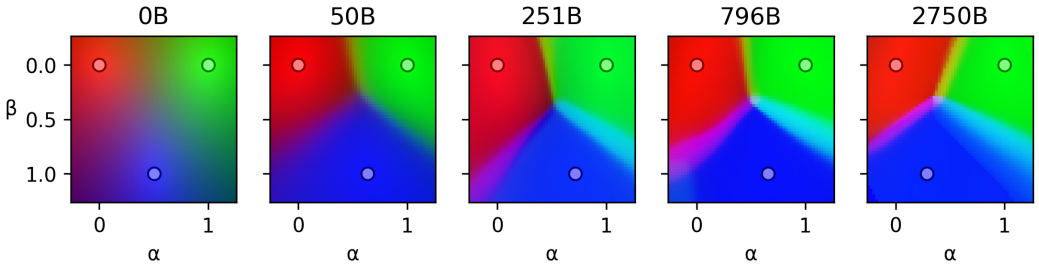

Figure 10

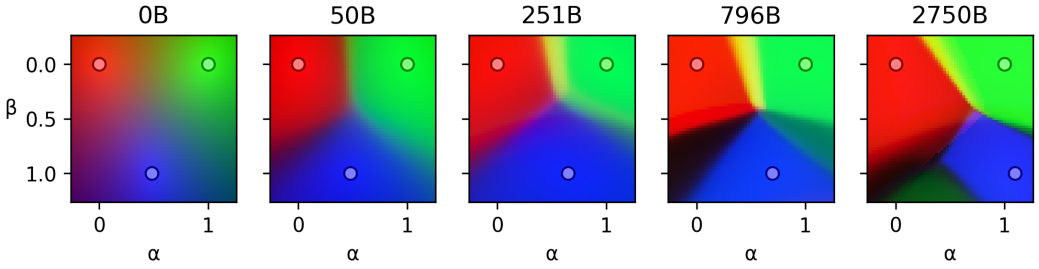

Figure 11